Prayas JEE 3.0 2025

Maths

Trigonometric Functions

DPP: 3

Q1 The maximum value of $1+\sin\left(\frac{\pi}{4}+\theta\right)+2\cos\left(\frac{\pi}{4}-\theta\right)$ for real values of θ is

(A) 3
(B) 5
(C) 4
(D) 2

Q2 The range of $f(\theta)=3\cos^2\theta-8\sqrt{3}\cos\theta\cdot\sin\theta+5\sin^2\theta-7$ is given by

(A) $[-7,7]$
(B) $[-10,4]$
(C) $[-4,7]$
(D) $[-10,7]$

Q3 If $A+B+C=180^\circ$, then $\frac{\sin 2A+\sin 2B+\sin 2C}{\cos A+\cos B+\cos C-1}$ is equal to

(A) $8\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$
(B) $8\cos\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$
(C) $8\sin\frac{A}{2}\cos\frac{B}{2}\cos\frac{C}{2}$
(D) $8\cos\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$

Q4 The value of the expression

$$y=\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\sqrt{\frac{1}{2}+\frac{1}{2}\cos 8\theta}}},$$

$$\text{where } \theta=\frac{\pi}{12}$$

simplifies to:

(A) 0
(B) 1
(C) an irrational number
(D) None of these

Q5 The minimum value of $27^{\cos 3x}\cdot 81^{\sin 3x}$ is

(A) 1
(B) $\frac{1}{81}$
(C) $\frac{1}{243}$
(D) $\frac{1}{27}$

Q6 The ratio of the greatest value of $2-\cos x+\sin^2 x$ is to its least value, is

(A) $7/4$
(B) $11/4$
(C) $13/4$
(D) none of these

Q7 The value of $\tan 10^\circ\cdot\tan 50^\circ\cdot\tan 70^\circ$ is

(A) $\sqrt{3}$
(B) $\frac{1}{\sqrt{3}}$
(C) 1
(D) -1

Q8 The value of $\tan\frac{\theta}{2}(1+\sec\theta)(1+\sec 2\theta)\left(1+\sec 2^2\theta\right)$ .... $1+\sec 2^n\theta)$ is

(A) $\tan 2^n\theta$
(B) $\tan 2^{n-1}\theta$
(C) $\tan 2^{n+1}\theta$
(D) $\tan 2^{n-2}\theta$

Q9 If $\theta=\frac{\pi}{2^n+1}$, then the value of $\cos\theta\cos 2\theta\cos 2^2\theta\ldots\cos 2^{n-1}\theta$ is

(A) 1
(B) $1/2^n$
(C) 2^n
(D) none of these

Q10 If $f(x)=2\sin^2\theta+4\cos(x+\theta)\sin x\sin\theta+\cos(2x+2\theta)$

then the value of $f^2(x) + f^2\left(\frac{\pi}{4} - x\right)$ is

(A) 0 (B) 1

(C) -1 (D) x^2

Answer Key

Q1	(C)	Q6	(C)
Q2	(B)	Q7	(B)
Q3	(B)	Q8	(A)
Q4	(C)	Q9	(B)
Q5	(C)	Q10	(B)